

Wedding Planning Automation System

Project Specification & Technical Blueprint

Prepared for: Internal Development Team

Client: Wedding Planning & Styling Business

Document type: Product & Technical Specification (v1.0)

Purpose: Single source of truth for building the MVP and full system

1. Executive Summary

The client runs a wedding planning and styling business and currently manages every wedding manually across four disconnected tools: Canva (mood boards / client-facing docs), Google Sheets (budgets, vendor lists, client data), Google Calendar (meetings, milestones), and Google Forms (client intake questionnaires). All of this is stitched together by hand by a human assistant, which does not scale past a handful of weddings and creates a high risk of missed deadlines, duplicated data entry, and things falling through the cracks.

This system replaces that patchwork with a single internal web application: one place to store every client, wedding, vendor and task, that automatically builds a planning timeline the moment a wedding date is entered, keeps Google Calendar in sync, generates and tracks tasks without manual input, and sends the right emails at the right time. The planner opens one dashboard each morning and immediately sees what needs attention — no more checking five different tabs.

1.1 Problem → Solution

Problem today	How the system solves it
Client and wedding data scattered across Sheets and Canva	Single structured database — one profile per client/wedding, always up to date
Planning timelines built by hand for every wedding	Timeline and milestones auto-generated from the wedding date on creation
Calendar events created manually for every meeting/milestone	Two-way Google Calendar integration schedules and updates events automatically
Intake forms filled out, then manually copied into Sheets/Canva	Client-facing forms write directly into the client's record — zero re-typing
Vendor contacts and quotes tracked loosely per wedding	Centralized, searchable vendor database linked to each wedding
Tasks and deadlines tracked in the planner's head or a notebook	Tasks auto-generated per wedding, assigned, tracked, and flagged when overdue
Reminder and follow-up emails written manually	Rule-based email automation for reminders, follow-ups, and information requests
No single view of what needs attention today	One dashboard: upcoming weddings, overdue tasks, today's meetings, at-a-glance status

2. Goals & Success Criteria

- The planner can open the dashboard each morning and know exactly what needs to happen that day, without opening Sheets, Canva, or Calendar.
- Creating a new wedding (client + date) automatically produces a full planning timeline and task list — no manual setup.
- Every client-facing intake form feeds directly into the client's record, with nothing re-typed by staff.
- Every wedding-related meeting appears on Google Calendar automatically and stays in sync if the date changes.
- The vendor database becomes the single place vendor details live, reusable across weddings.

- Routine emails (reminders, follow-ups, requests for info) go out automatically based on the timeline, not from memory.

3. Scope — MVP vs. Full System

The client explicitly asked to start with an MVP. The table below separates what ships first from what follows in later phases, matching the client's own priority list from the job post.

Module	MVP (Phase 1)	Later phase
Client & Wedding DB	Full CRUD, profiles, budgets, venues, notes, status	—
Wedding Dashboard	Upcoming weddings, overdue tasks, today's meetings	Custom widgets, saved views
Automated Planning Timeline	Rule-based milestone generation from wedding date	Editable timeline templates per package type
Task Generation	Auto-create standard task list per wedding	Smart re-prioritization, workload balancing
Google Calendar Integration	One-way sync: system → Calendar for meetings/milestones	Two-way sync, conflict detection
Client Info Collection	Web form → writes directly into client record	Branded/multi-step client portal
Vendor Database	CRUD, categories, link vendors to weddings	Vendor portal, quote comparison
Email/Task Automation	Templated reminders + follow-ups on a schedule	Two-way email thread tracking, smart replies
Canva integration	Not in MVP (manual export still fine)	Auto-push mood boards / auto-fill Canva templates

Everything in the MVP column maps 1:1 to the bullet list the client posted in the job description, so the MVP alone is enough to demonstrate the full value proposition to the client early.

4. System Architecture

A standard three-tier web app: a React frontend, a REST API backend, a relational database, plus two external integrations (Google Calendar API and an email-sending service). Kept deliberately simple — this is an internal tool for one small business, not a multi-tenant SaaS.

4.1 Recommended Stack

Layer	Choice	Why
Frontend	Next.js (React) + TypeScript + Tailwind CSS	Fast to build a dashboard-style app; strong ecosystem; matches the 'Next.js/React' skills the client listed
Backend / API	Node.js + Express (or FastAPI/Python)	Straightforward REST API; both are on the client's mandatory skills list — pick whichever matches the developer's strength
Database	PostgreSQL	Relational data (clients ↔ weddings ↔ tasks ↔ vendors) fits a relational model much better than a spreadsheet or NoSQL store
Auth	Simple email/password or Google OAuth (single planner + maybe 1 assistant login)	Only 1–2 internal users; no need for complex role systems in v1
Calendar	Google Calendar API	Client already lives in Google Calendar — sync into it rather than replace it
Forms	Custom form pages inside the app (or Typeform/Tally webhook) → API	Removes the manual copy-paste step from Google Forms into Sheets
Email	Gmail API or a transactional service (Resend / SendGrid)	Automated, template-driven sends triggered by the timeline/task engine
Hosting	Vercel (frontend) + Railway/Render (API + Postgres)	Cheap, fast to deploy, easy for a solo/small-team build, good portfolio story

4.2 High-Level Data Flow

- Planner creates a Client → creates a Wedding for that client with a date → Timeline Engine generates milestones + tasks automatically.
- Timeline Engine pushes meeting/milestone events to Google Calendar via the Calendar API.
- Client fills an intake Form on a public link → data lands directly on that Wedding record.
- Planner adds Vendors to a Wedding from the shared Vendor Database (or creates a new vendor inline).
- A scheduled job (cron) checks the Timeline daily → sends templated emails for anything due, and flags overdue tasks on the Dashboard.
- Dashboard queries all of the above and renders one unified 'what needs attention today' view.

5. Data Model

Core entities and their key fields. This maps directly to the database tables/API resources.

Client

Field	Type	Notes
id	UUID	Primary key
full_name / partner_name	text	Both partners, optional second name
email / phone	text	Primary contact channel for automated emails
status	enum	lead / active / completed / archived
notes	text	Free-form planner notes
created_at	timestamp	—

Wedding

Field	Type	Notes
id	UUID	Primary key
client_id	FK → Client	One client can have one wedding record (or more, for repeat clients)
wedding_date	date	Drives the whole timeline engine
venue	text / FK → Venue	Can start as text, normalize later
budget_total / budget_spent	decimal	Simple running budget tracker
planning_status	enum	inquiry / booked / in-progress / final-month / completed
style_notes	text	Mood board / styling brief notes

Task

Field	Type	Notes
id	UUID	Primary key
wedding_id	FK → Wedding	Every task belongs to one wedding
title / description	text	e.g. 'Confirm florist final order'
due_date	date	Auto-calculated relative to wedding_date
status	enum	todo / in-progress / done / overdue (derived)
priority	enum	low / medium / high
assignee	text / FK → User	Planner or assistant
source	enum	auto-generated / manual

Vendor

Field	Type	Notes
id	UUID	Primary key
name / category	text / enum	e.g. florist, caterer, photographer, venue, DJ
contact_email / phone	text	Used by email automation
notes / pricing_notes	text	Quotes, past experience
linked_weddings	M:N → Wedding	A vendor can serve many weddings

Meeting / CalendarEvent

Field	Type	Notes
id	UUID	Primary key
wedding_id	FK → Wedding	—
type	enum	milestone / client-meeting / vendor-meeting / reminder
scheduled_at	timestamp	—
google_event_id	text	Stored after pushing to Google Calendar, used to update/cancel it later

6. Feature Breakdown

6.1 Client & Wedding Management

- Create/edit/archive client profiles: names, contact info, referral source, status.
- Each client can have a linked Wedding record: date, venue, budget, planning status, style notes.
- Searchable/filterable client + wedding list (by status, date range, upcoming vs. past).
- A single 'Wedding Detail' page acts as the hub: client info, timeline, tasks, vendors, budget, notes — everything for that wedding in one screen.

6.2 Automated Planning Timeline

- On wedding creation (or whenever wedding_date changes), the system generates a milestone timeline working backward from the date — e.g. 12 months out: book venue; 9 months: book major vendors; 3 months: final headcount; 1 month: final walkthrough; 1 week: confirm all vendors.
- Timeline rules live in a config table (not hardcoded), so the planner or dev can adjust the standard milestone template without touching code.
- Each milestone can spawn one or more Tasks and, optionally, a Calendar meeting.

6.3 Google Calendar Integration

- OAuth connection to the planner's Google account (one-time setup).
- Every milestone/meeting created in-app is pushed to a dedicated 'Weddings' calendar automatically.
- If a wedding date or a task due date changes, the corresponding calendar event updates (not duplicates).
- MVP: one-way sync (app → Calendar). Two-way sync (Calendar edits reflected back) is a fast-follow, not MVP.

6.4 Client Forms

- Simple public intake form (e.g. /forms/intake/[wedding_id]) the planner sends to each couple.
- Submission writes straight into that Wedding's record — style preferences, guest count, key dates, contact details — no manual transcription into Sheets or Canva.
- Optional: export a submission as a formatted brief the planner can drop into Canva.

6.5 Vendor Database

- Central vendor list, categorized (florist, caterer, venue, photographer, DJ/band, hair & makeup, etc.).
- Vendors can be linked to one or more weddings, with per-wedding notes (quoted price, confirmed status).
- Quick 'vendors contacted for this wedding' view replaces the manual Sheet the client described.

6.6 Task Management

- Tasks auto-generated from the timeline template, editable and re-assignable.
- Manual tasks can also be added ad hoc per wedding.
- Status tracking (todo / in-progress / done) with automatic 'overdue' flag once due_date passes.
- Tasks roll up into the Dashboard's 'needs attention' list.

6.7 Email Automation

- Template library: client reminders, vendor follow-ups, information requests, milestone confirmations.

- A daily scheduled job checks the timeline/tasks and sends the right templated email automatically (e.g. '30 days out' reminder, 'vendor hasn't confirmed' follow-up).
- Sent emails are logged against the wedding record so the planner can see what's already gone out.

6.8 Dashboard

- Upcoming weddings (sorted by date, with a countdown).
- Overdue and due-soon tasks across all weddings.
- Today's / this week's meetings pulled from the calendar integration.
- Weddings needing attention — flagged automatically (e.g. no vendor confirmed 60 days out).
- This is the screen the planner opens first every morning, per the client's stated goal.

7. Key User Flows

7.1 Onboarding a new wedding

- Planner clicks 'New Client' → enters names + contact info.
- Planner adds a Wedding to that client → enters wedding_date + venue (if known).
- System auto-generates the milestone timeline and initial task list.
- System pushes milestone meetings to Google Calendar.
- Planner sends the client the intake form link (auto-generated per wedding).
- Client fills the form → data lands in the Wedding record automatically.

7.2 Daily planner workflow

- Planner opens the Dashboard.
- Sees overdue tasks across all weddings, today's meetings, and weddings needing attention.
- Clicks into a flagged wedding → reviews timeline, updates a task status, or adds a vendor.
- Any changes automatically re-sync to Calendar and re-evaluate the Dashboard's flags.

7.3 Vendor management

- Planner searches the Vendor Database for a category (e.g. 'florist').
- Links an existing vendor to the current wedding, or adds a new one inline.
- Marks vendor status ('contacted', 'quoted', 'confirmed') per wedding.

8. Delivery Plan

Matches the client's stated 1–3 month duration and 30+ hrs/week availability expectation.

Phase	Duration	Deliverables
Phase 0 — Setup	2–3 days	Repo, hosting, DB schema, auth, Google OAuth app registration
Phase 1 — Core data	1 week	Client & Wedding CRUD, Vendor DB, basic UI shell
Phase 2 — Automation engine	1–2 weeks	Timeline generator, auto task creation, budget tracking
Phase 3 — Integrations	1 week	Google Calendar sync, intake form → record, email templates + scheduler
Phase 4 — Dashboard & polish	3–5 days	Unified dashboard, flags/alerts, UI polish, QA
Phase 5 — MVP handoff	2–3 days	Deploy, client walkthrough, documentation, feedback loop

9. Out of Scope (v1)

- Direct Canva API integration (manual export/import remains fine for MVP).
- Multi-tenant support for other planning businesses — this is a single-business internal tool.
- Client-facing payment processing / invoicing.
- Mobile app — a responsive web app covers the client's needs.

10. Deliverables Checklist for Proposal

Per the client's job post, a proposal should explicitly include:

- Examples of similar systems previously built (portfolio links/screenshots).
- Estimated MVP timeframe.
- Estimated budget or hourly rate.
- Availability for ongoing development after MVP.
- A short description of recent experience with similar (CRM/automation-style) projects.